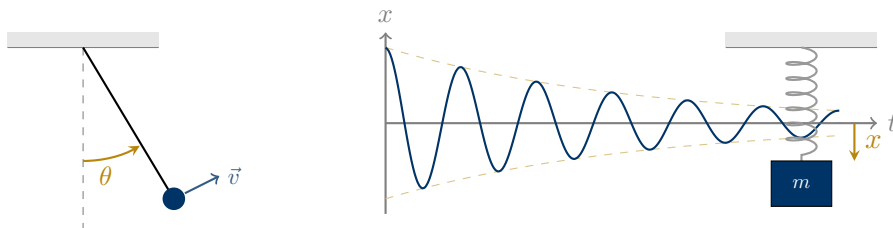

Oscillations

Selected Problems & Solutions to Kevin Zhou's Handouts



Haoenchan

haoenphysics

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1 Hole

Problem 1: Hole in an Infinite Sheet (Morin 5.13)

A hole of radius R is cut out from an infinite flat sheet with mass per unit area σ . Let L be the line that is perpendicular to the sheet and passes through the center of the hole.

- (a) What is the force on a mass m located on L at a distance x from the center?
- (b) If released from rest at $x \ll R$, find the approximate angular frequency of oscillation.
- (c) If $x \gg R$, find the period of oscillation.

Solution:

(a) Consider a ring within the sheet with radius r and thickness dr . Its mass is $dM = 2\pi r\sigma dr$. By symmetry, only the force component along L survives. The contribution is:

$$dF = -\frac{Gm dM}{x^2 + r^2} \cos \theta = -\frac{Gm(2\pi r\sigma dr)}{x^2 + r^2} \frac{x}{\sqrt{x^2 + r^2}}$$

Integrating from R to ∞ :

$$F = -2\pi Gm\sigma x \int_R^\infty \frac{r dr}{(x^2 + r^2)^{3/2}} = -2\pi Gm\sigma x \left[\frac{-1}{\sqrt{x^2 + r^2}} \right]_R^\infty = -\frac{2\pi Gm\sigma x}{\sqrt{x^2 + R^2}}$$

(b) For $x \ll R$, we approximate the denominator: $\sqrt{x^2 + R^2} \approx R$.

$$F \approx -\frac{2\pi G\sigma m}{R} x$$

This is the form of Hooke's Law $F = -k_{eff}x$. The frequency is:

$$\omega = \sqrt{\frac{k_{eff}}{m}} = \sqrt{\frac{2\pi G\sigma}{R}}$$

(c) For $x \gg R$, the force approaches a constant (similar to an infinite sheet without a hole):

$$F \approx -2\pi Gm\sigma$$

This corresponds to a uniform acceleration $g_{eff} = 2\pi G\sigma$. The time to fall from height x is $t_{fall} = \sqrt{2x/g_{eff}}$. Since the period is $4 \times t_{fall}$:

$$T = 4\sqrt{\frac{x}{\pi G\sigma}}$$

2 Energy

Problem 8: Mass on a Plane (Cahn)

A particle of mass M moves on a frictionless horizontal plane. It is connected by a massless string passing through a hole to a second mass m hanging vertically. M moves in a circle of radius r while m remains stationary. Find the frequency of small oscillations in r .

Solution: Let r be the generalized coordinate. The length of the string is fixed, so if M is at radius r , m is at height $-r$ (up to a constant).

Kinetic Energy: Both masses move with radial speed $\dot{r}$. M also has tangential velocity $v_\theta = L/(Mr)$ where L is angular momentum.

$$K = \frac{1}{2}(M + m)\dot{r}^2 + \frac{L^2}{2Mr^2}$$

Potential Energy:

$$V = mgr$$

The effective potential for the radial motion includes the centrifugal barrier:

$$V_{eff}(r) = mgr + \frac{L^2}{2Mr^2}$$

To find the equilibrium r_0 , we set $V'_{eff}(r_0) = 0$:

$$mg - \frac{L^2}{Mr_0^3} = 0 \implies \frac{L^2}{Mr_0^3} = mg$$

To find the frequency, we compute the second derivative (the effective spring constant):

$$k_{eff} = V''_{eff}(r_0) = \frac{3L^2}{Mr_0^4} = \frac{3}{r_0} \left(\frac{L^2}{Mr_0^3} \right) = \frac{3mg}{r_0}$$

The effective mass for the radial oscillation is $m_{eff} = M + m$ (since both masses must accelerate radially).

$$\omega = \sqrt{\frac{k_{eff}}{m_{eff}}} = \sqrt{\frac{3mg/r_0}{M + m}} = \sqrt{\frac{3g}{r_0} \frac{m}{M + m}}$$

3 Continuous Systems

Problem 12: The Massive Spring

A uniform spring of constant k and mass m is attached to a wall and a mass M . (a) Show

that for $m \ll M$, the frequency is $\omega = \sqrt{k/(M + m/3)}$. (b) Generalize for arbitrary mass ratios.

Solution: (a) Assume the spring stretches uniformly. If the end mass M moves with velocity v_0 , a point on the spring at distance x (where total length is L) moves at $v(x) = v_0(x/L)$. The kinetic energy of the spring is:

$$K_{spring} = \int_0^L \frac{1}{2}(\rho dx) \left(v_0 \frac{x}{L}\right)^2 = \frac{1}{2} \frac{m}{L} \frac{v_0^2}{L^2} \left[\frac{x^3}{3}\right]_0^L = \frac{1}{6} m v_0^2$$

Total Kinetic Energy:

$$K_{tot} = \frac{1}{2} M v_0^2 + \frac{1}{6} m v_0^2 = \frac{1}{2} \left(M + \frac{m}{3}\right) v_0^2$$

Thus $m_{eff} = M + m/3$.

(b) For the general case, we treat the spring as a wave. The displacement $u(x, t)$ satisfies the wave equation. The boundary condition at the mass M ($x = L$) is:

$$-F_{spring} = M\ddot{u}(L, t) \implies -kL \frac{\partial u}{\partial x} = M\ddot{u}$$

Assuming a standing wave solution $u(x, t) = \sin(qx) \cos(\omega t)$ where $\omega = v_{wave}q$:

$$-kLq \cos(qL) = -M\omega^2 \sin(qL)$$

Let $\alpha = \sqrt{m/M}$. The condition simplifies to the transcendental equation:

$$\tan(\alpha\omega/\omega_0) = \frac{\alpha\omega_0}{\omega} \quad \text{where } \omega_0 = \sqrt{k/M}$$

4 Coupled Oscillators & Normal Modes

Problem 21: Beats (Morin 4.10)

Three springs and two equal masses lie between two walls. The outer springs have constant k , and the middle spring has constant $\kappa \ll k$. If $x_1(0) = a$ and $x_2(0) = 0$, describe the motion.

Solution: The equations of motion are:

$$m\ddot{x}_1 = -kx_1 - \kappa(x_1 - x_2)$$

$$m\ddot{x}_2 = -kx_2 - \kappa(x_2 - x_1)$$

We switch to normal coordinates (sum and difference):

- Let $y_1 = x_1 + x_2$ (Center of Mass mode). Adding the EOMs:

$$m\ddot{y}_1 = -ky_1 \implies \omega_1 = \sqrt{k/m}$$

- Let $y_2 = x_1 - x_2$ (Relative mode). Subtracting the EOMs:

$$m\ddot{y}_2 = -(k + 2\kappa)y_2 \implies \omega_2 = \sqrt{(k + 2\kappa)/m}$$

Initial conditions imply $y_1(0) = a$ and $y_2(0) = a$. Reconstructing the positions:

$$x_1(t) = \frac{1}{2}(y_1 + y_2) = \frac{a}{2}(\cos(\omega_1 t) + \cos(\omega_2 t))$$

Using sum-to-product identities and defining $\epsilon \approx \omega_1(\kappa/2k)$:

$$x_1(t) \approx a \cos((\omega + \epsilon)t) \cos(\epsilon t)$$

This represents **beats**: the energy sloshes back and forth between the two masses at the slow frequency ϵ .

Problem 23: Masses on a Hoop (IPhO 1986)

N identical masses m are on a circular hoop connected by springs k . Find the normal modes.

Solution: This system has discrete translational symmetry ($j \rightarrow j+1$). We guess a solution of the form of a traveling wave around the ring:

$$x_j(t) = A e^{i(\omega t + j\phi)}$$

The boundary condition is the circle: $x_{N+1} = x_1$, which implies $e^{iN\phi} = 1$. Thus the phase shift between neighbors must be $\phi_n = \frac{2\pi n}{N}$ for $n = 0, \dots, N-1$.

The equation of motion for mass j is:

$$m\ddot{x}_j = -k(x_j - x_{j-1}) - k(x_j - x_{j+1}) = -k(2x_j - x_{j-1} - x_{j+1})$$

Plugging in the ansatz:

$$-m\omega^2 = -k(2 - e^{-i\phi_n} - e^{i\phi_n}) = -k(2 - 2\cos\phi_n)$$

$$\omega_n^2 = \frac{2k}{m}(1 - \cos\phi_n) = \frac{4k}{m}\sin^2\left(\frac{\phi_n}{2}\right)$$

$$\omega_n = 2\sqrt{\frac{k}{m}} \left| \sin\left(\frac{\pi n}{N}\right) \right|$$

5 Adiabatic Invariants

Problem 25: Adiabatic Pendulum

Consider a pendulum whose length slowly changes from L to $L/2$. How does the amplitude θ_0 change?

Solution: For a system with a slowly varying parameter, the action $J = \oint p dq$ is an adiabatic invariant (conserved). For a pendulum, $E = \frac{1}{2}mL^2\dot{\theta}^2 + \frac{1}{2}mgL\theta^2$. This traces an ellipse in phase space (p_θ vs θ). The area of this ellipse is proportional to Energy / Frequency:

$$J \propto \frac{E}{\omega}$$

Since $E \propto mgL\theta_0^2$ and $\omega \propto \sqrt{g/L}$:

$$\frac{mgL\theta_0^2}{\sqrt{g/L}} = \text{constant} \implies \theta_0^2 L^{3/2} = \text{constant}$$

$$\theta_0 \propto L^{-3/4}$$

If $L \rightarrow L/2$, the new amplitude is $\theta_{new} = \theta_0(2)^{3/4} \approx 1.68\theta_0$. Unlike the intuition in Poe's *The Pit and the Pendulum*, lengthening the pendulum decreases the amplitude, while shortening it increases the amplitude.

6 Buoyancy and Generalized Forces

Problem 2: Oscillating Floating Objects

- (a) A cubical glacier of side length L and density ρ_i floats in water of density ρ_w . Find the angular frequency of small vertical oscillations.
- (b) A ball of radius R floats in water with exactly half its volume submerged. Find the frequency of small oscillations.

Solution: (a) Let x be the vertical displacement from equilibrium. The restoring force is the change in buoyant force:

$$F_{restore} = -(\rho_w g)(\text{Area})x = -\rho_w g L^2 x$$

The total mass of the glacier is $M = \rho_i L^3$. Using Newton's Second Law:

$$(\rho_i L^3)\ddot{x} = -\rho_w L^2 g x \implies \omega = \sqrt{\frac{\rho_w L^2 g}{\rho_i L^3}} = \sqrt{\frac{\rho_w g}{\rho_i L}}$$

(b) Since the ball is half submerged, its density must be $\rho_{ball} = \frac{1}{2}\rho_w$. The effective spring constant comes from the cross-sectional area at the water line, which is a circle of radius R :

$$k = \rho_w g (\pi R^2)$$

The mass of the ball is $m = \rho_{ball} V = \frac{1}{2}\rho_w (\frac{4}{3}\pi R^3) = \frac{2}{3}\pi \rho_w R^3$.

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{\pi R^2 \rho_w g}{(\frac{2}{3}\pi \rho_w R^3)}} = \sqrt{\frac{3g}{2R}}$$

Note: This neglects the "virtual mass" of the water that moves with the object.

Problem 6: Rope in a Tube

A rope of length l and uniform mass per length λ lies inside a frictionless tube. The tube is curved arbitrarily, but the left end is higher than the right end by a vertical height h . If released from rest, find the acceleration.

Solution: Instead of resolving tension forces along the curve, use **Generalized Coordinates**. Let q be the distance the rope has slid along the tube.

Kinetic Energy: Every part of the rope moves at speed $\dot{q}$.

$$K = \frac{1}{2} M_{total} \dot{q}^2 = \frac{1}{2} (\lambda l) \dot{q}^2$$

Generalized Force: When the rope moves by dq , a small chunk of mass $dm = \lambda dq$ is effectively transferred from the back (top) to the front (bottom). The work done by gravity is the change in potential energy:

$$dV = -(dm)gh = -(\lambda dq)gh$$

Thus, the generalized force is $F_q = -\frac{\partial V}{\partial q} = \lambda gh$.

Using the energy equation $\frac{d}{dt}(K) = \text{Power}$:

$$\frac{d}{dt} \left(\frac{1}{2} \lambda l \dot{q}^2 \right) = (\lambda gh) \dot{q}$$

$$\lambda l \dot{q} \ddot{q} = \lambda gh \dot{q} \implies \ddot{q} = \frac{gh}{l}$$

7 Damped and Driven Systems

Problem 17: The Driven Oscillator

A damped harmonic oscillator obeys the equation:

$$m\ddot{x} + b\dot{x} + kx = F_0 \cos(\omega t)$$

Using complex exponentials, find the steady-state amplitude A_0 and phase shift ϕ .

Solution: We complexify the driving force to $F_0 e^{i\omega t}$ and look for a solution $x(t) = \tilde{A} e^{i\omega t}$. Plugging this into the differential equation:

$$m(i\omega)^2 \tilde{A} + b(i\omega) \tilde{A} + k \tilde{A} = F_0$$

$$\tilde{A}(-m\omega^2 + ib\omega + k) = F_0$$

$$\tilde{A} = \frac{F_0}{(k - m\omega^2) + i(b\omega)}$$

The physical amplitude A_0 is the magnitude $|\tilde{A}|$:

$$A_0 = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (b\omega)^2}}$$

The phase shift ϕ is the argument of the complex number in the denominator:

$$\tan \phi = \frac{b\omega}{k - m\omega^2}$$

8 Springs

Problem 10: Vector Springs (Morin 4.20)

A mass m is attached to n springs with relaxed lengths of zero and spring constants $k_1, k_2, \dots, k_n$. The anchor points are at arbitrary positions $\mathbf{r}_i$. If the mass is given a kick from equilibrium, describe the motion.

Solution: The force from the i -th spring is $\mathbf{F}_i = -k_i(\mathbf{r} - \mathbf{r}_i)$. The total force is:

$$\mathbf{F}_{net} = \sum_i -k_i(\mathbf{r} - \mathbf{r}_i) = -\left(\sum_i k_i\right)\mathbf{r} + \sum_i k_i\mathbf{r}_i$$

This is of the form $\mathbf{F} = -K\mathbf{r} + \mathbf{C}$, where $K = \sum k_i$ and $\mathbf{C}$ is a constant vector. This describes Simple Harmonic Motion about a shifted equilibrium point with angular frequency:

$$\omega = \sqrt{\frac{\sum k_i}{m}}$$

Problem 11: Projectile on a Spring (Morin 4.22)

A projectile of mass m is attached to a zero-length spring of constant k . It is thrown with velocity v at an angle θ . (a) What is the shape of the trajectory? (b) Find the velocity v such that the projectile hits the ground traveling straight downward.

Solution: (a) The force is $\mathbf{F} = -k\mathbf{r} - mg\hat{y}$. This is equivalent to a single spring force centered at a shifted equilibrium $\mathbf{r}_{eq} = (0, -mg/k)$. The x and y motions are independent SHM with the same $\omega = \sqrt{k/m}$. Thus, the trajectory is an **ellipse**.

(b) Let the vertical velocity be $v_y(t) = v_0 \cos(\omega t + \phi)$. We need the landing time to correspond to a phase change of $\pi/2$ (a quarter cycle) so that horizontal velocity vanishes (if started at max) or similar. Specifically, hitting the ground straight down means $v_x = 0$. Since $v_x(t) = v \cos \theta \cos(\omega t)$, this happens at $\omega t = \pi/2$. At this time, we need $y = 0$. The

vertical solution involves a phase shift $\phi = \pi/4$. Solving the kinematic constraints yields:

$$v = \sqrt{\frac{m}{k} \frac{g}{\sin \theta}}$$

9 PhO Problems, Solutions

IPhO 1984, Problem 2

Estimate the period of oscillation for water sloshing back and forth (seiching) in a rectangular container of length L and water depth h , where $h \ll L$. Use energy methods.

Solution: We approximate the motion by considering the water surface tilting by a small amount ξ .

Potential Energy: The water displacement moves a triangular wedge of water from one side to the other. The mass of this wedge is $m = \frac{1}{2}(L/2)(\xi w)\rho$, where w is the width. The center of mass of this wedge rises by a distance $2\xi/3$ relative to the lowered side. The total potential energy change is:

$$U \approx mg \left(\frac{2\xi}{3} \right) = \frac{1}{6} \rho L w g \xi^2$$

Kinetic Energy: We estimate the kinetic energy by tracking the center of mass of the entire water body. The center of mass shifts horizontally by Δx_{cm} . Using the moment of the displaced wedge:

$$M \Delta x_{cm} = m \left(\frac{2L}{3} \right) \implies \Delta x_{cm} = \frac{1}{6} \frac{L \xi}{h}$$

The total mass is $M = \rho L w h$. The kinetic energy is approximately:

$$K \approx \frac{1}{2} M \dot{x}_{cm}^2 = \frac{1}{2} (\rho L w h) \left(\frac{L \dot{\xi}}{6h} \right)^2 = \frac{\rho w L^3}{72h} \dot{\xi}^2$$

Period: Equating the energy forms to a harmonic oscillator ($E = \frac{1}{2} m_{eff} \dot{\xi}^2 + \frac{1}{2} k_{eff} \xi^2$), we find:

$$\omega^2 = \frac{k_{eff}}{m_{eff}} = \frac{\rho L w g / 3}{\rho w L^3 / 36h} = \frac{12gh}{L^2}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{L^2}{12gh}}$$

Note: This result agrees dimensionally with the shallow water wave speed $v = \sqrt{gh}$.

PPP 79 (Problem 4)

A pendulum of length L hangs in a train. The train moves with uniform acceleration a . Find the period of small oscillations.

Solution: In the train's non-inertial frame, there is a fictitious force $-ma$. This combines with gravity to create an effective gravitational field:

$$\vec{g}_{eff} = -a\hat{x} - g\hat{y} \implies g_{eff} = \sqrt{g^2 + a^2}$$

The period for small oscillations is:

$$T = 2\pi\sqrt{\frac{L}{g_{eff}}} = \frac{2\pi\sqrt{L}}{(g^2 + a^2)^{1/4}}$$

PPP 77 (Problem 13)

A small bob of mass m is attached to two light, unstretched, identical springs of length L . If displaced perpendicular to the line by l , the period is T . Find the period if displaced by $2l$.

Solution: For a transverse displacement x , the stretch is $\Delta L \approx x^2/2L$. The tension is $F_{tens} = k(x^2/2L)$. The restoring force component is $F_{restoring} \approx -2F_{tens}(x/L) \propto -x^3$. For a restoring force $F \propto x^3$, the potential is $V \propto x^4$. Dimensional analysis for period T with potential $V = Cx^n$ gives $T \propto A^{(2-n)/2}$. Here $n = 4$, so $T \propto A^{-1}$. Doubling the amplitude ($2l$) halves the period.

$$T_{new} = T/2$$

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A uniform rod of length $2L$ hangs from the ceiling by two vertical massless strings of length L attached to its ends. The rod is pulled a small distance horizontally and released (remaining parallel to the ground). Find the period of oscillation.

Solution: Let θ be the angle the *strings* make with the vertical. This is the generalized coordinate. Since the rod remains horizontal, it does not rotate about its center of mass, so there is no rotational kinetic energy term ($\frac{1}{2}I\omega^2 = 0$). The center of mass moves exactly like a simple pendulum of length L :

$$K = \frac{1}{2}M(L\dot{\theta})^2$$

$$V = MgL(1 - \cos\theta) \approx \frac{1}{2}MgL\theta^2$$

This is mathematically identical to a simple pendulum of length L .

$$T = 2\pi\sqrt{\frac{L}{g}}$$